ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 1 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 A ✓
1.2 C ✓
1.3 B ✓
1.4 D ✓
1.5 A ✓
1.6 A ✓
1.7 C ✓
1.8 C ✓
1.9 B ✓
1.10 B ✓

(10 × 1) [10]

QUESTION 2

2.1 True ✓
2.2 True ✓
2.3 True ✓
2.4 False ✓
2.5 False ✓
2.6 True ✓
2.7 False ✓
2.8 True ✓
2.9 True ✓
2.10 False ✓

(10 × 1) [10]

QUESTION 3

3.1 C ✓
3.2 G ✓
3.3 I ✓
3.4 E ✓
3.5 A ✓
3.6 J ✓
3.7 H ✓

3.8 F ✓

3.9 B ✓

3.10 D ✓

(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1.1 $V_B = V_{CC} \times R_2 \div (R_1 + R_2)$ ✓ $= 12 \times 10 \div 57 = 2{,}11\ V$ ✓ (2)

4.1.2 $V_E = 2{,}11 - 0{,}7 = 1{,}41\ V$ ✓ (1)

4.1.3 $I_E = V_E \div R_E$ ✓ $= 1{,}41 \div 1\,000 = 1{,}41\ mA$ ✓ (2)

4.1.4 $V_C = V_{CC} - I_C R_C$ ✓ $= 12 - 1{,}41 \times 10^{-3} \times 3\,300 = 7{,}36\ V$ ✓ (accept 7,34–7,37 V) (2)

4.2 The output is 180° out of phase with the input (inverted) ✓. Coupling capacitors pass the AC signal ✓ but block DC, so the signal source and the load do not upset the bias (Q-point) ✓. (3)

4.3 CE – voltage amplifier / general-purpose amplifier stage ✓; CC (emitter follower) – buffer / impedance matching / driving a low-resistance load ✓; CB – high-frequency (RF) amplifier / current buffer ✓ (3)

4.4 As a transistor heats up its collector current rises, which heats it more, until it is destroyed ✓✓. With R_E, a rise in I_E raises V_E and lowers V_{BE}, which reduces the base current (negative feedback) and stabilises the current ✓. (3)

[16]

QUESTION 5

5.1 $A_v = 1 + R_f \div R_i = 1 + 90 \div 10 = 10$ ✓✓; $V_{out} = 10 \times 0{,}25 = 2{,}5\ V$ ✓ (3)

5.2 $V_{out} = -R_f(V_1 \div R_1 + V_2 \div R_2)$ ✓ $= -10\ (1 \div 10 + 2 \div 10)$ ✓ $= -3\ V$ ✓ (3)

5.3.1 $\text{Resolution} = V_{ref} \div (2^{10} - 1)$ ✓ $= 5 \div 1\,023 = 4{,}89\ mV$ ✓ ($5 \div 1\,024 = 4{,}88\ mV$ accepted) (2)

5.3.2 $D = 2 \div 5 \times 1\,023$ ✓ $= 409$ ✓ (accept 409 or 410) (2)

5.4 The number of samples (conversions) taken per second ✓. It must be at least twice the highest frequency in the signal (Nyquist) ✓, otherwise fast changes are missed and the signal is distorted (aliasing) ✓. (3)

5.5 $D = 1011 = 11$ ✓; $V_{out} = 11 \div 16 \times 5$ ✓ $= 3{,}44\ V$ ✓ (3)

5.6 It has a very high input impedance, so it does not load the sensor ✓, and a low output impedance, so it can drive the ADC input without changing the voltage ✓. (2)

[18]

QUESTION 6

6.1 Table ✓✓: on the active clock edge Q takes the value of D (D = 0 → Q = 0; D = 1 → Q = 1); without a clock edge Q does not change. Explanation ✓✓: the D input is only copied to the output on the clock edge, so it stores (latches) one bit until the next clock pulse. (4)

6.2 Asynchronous: the clock drives only the first flip-flop and each flip-flop is clocked by the output of the one before ✓, so changes ripple through with small delays and the counter is slower ✓. Synchronous: all flip-flops share the same clock and change at the same time ✓, so it is faster and has no ripple errors ✓. (4)

6.3 4 flip-flops ✓: 3 flip-flops give only 8 states, 4 give 16 states, which is enough for 12 (with reset at 12) ✓ (2)

6.4 After clock 1: 0101 ✓✓; after clock 2: 0010 ✓ (3)

6.5 Any TWO ✓✓: serial-to-parallel conversion to drive many LEDs or relays from few pins (e.g. 74HC595); reading many buttons / sensors (parallel-to-serial); delaying or storing data; SPI communication. (2)

6.6 When a mechanical switch closes, its contacts bounce and make and break several times in a few milliseconds ✓, so a digital circuit counts many presses ✓. Removed with an RC filter and Schmitt trigger, an SR latch, or a software delay / debounce routine ✓. (3)

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QUESTION 7

7.1 The strain gauge is one arm of the bridge, which is balanced (zero output) with no load ✓; a force stretches the gauge and changes its resistance slightly ✓, unbalancing the bridge and giving a small output voltage that is amplified and is proportional to the force ✓. (3)

7.2 The thermistor's resistance falls as the temperature rises ✓, so the share of the supply voltage across it (or across the fixed resistor) changes and gives a voltage that represents the temperature ✓. (2)

7.3 Any FOUR ✓✓✓✓: amplification; filtering (noise removal); isolation; level shifting / scaling; linearisation; conversion (e.g. resistance to voltage, current to voltage); analogue-to-digital conversion. (4)

7.4 A positive pulse on the gate (with the anode positive) switches the SCR on ✓, and it stays on even when the gate pulse is removed ✓. On AC it switches off when the current falls to zero at the end of each half-cycle (below the holding current) ✓. (3)

7.5 $V_{avg} = 0{,}4 \times 12$ ✓ $= 4{,}8$ V ✓; $T = 1 \div 1\,000 = 1$ ms ✓; $t_{on} = 0{,}4 \times 1$ ms $= 0{,}4$ ms ✓ (4)

7.6 It passes the control signal as light, so there is no electrical connection ✓, protecting the microcontroller from high voltages, noise and spikes from the motor circuit ✓. (2)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE